

NIHAL POTHUNOORI

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EDUCATION

University of Illinois at Urbana-Champaign

Expected May 2027

B.S. in Electrical Engineering

Relevant Coursework: Control Systems; Principles of Safe Autonomy; Analog Signal Processing; Computer Systems & Programming

SKILLS

Programming: Python, C/C++, Matlab, Java, C#, .NET

Frameworks: ROS2, OpenCV, VLAs, PyTorch, Deep Learning, Reinforcement Learning

EXPERIENCE

DASLABS, University of Illinois

Jan 2026 - Present

Undergraduate Researcher

Champaign, IL

- Implementing vision-language-action (VLA) models to identify stable rest/anchor locations for a soft robot during locomotion and manipulation tasks in confined space.
- Building a ROS2-based multimodal perception + evaluation pipeline in Python (RGB-D segmentation, logging, metrics) to benchmark rest-point selection reliability.

College of Education, University of Illinois

Aug 2025 – Present

Web Developer/IT Support

Champaign, IL

- Built a .NET 8 C# library integrating with Azure Active Directory and automated permissions management for ~2,500 users; shipped web updates in HTML/C#.

Autonomous Materials Systems Group, University of Illinois

Aug 2025 – Jan 2026

Undergraduate Researcher

Champaign, IL

- Designed and built a robotic manipulator + gantry system for automated DCPD polymer sample preparation in a DARPA-funded lab project.
- Applied PyTorch-based reinforcement learning for closed-loop process control; improved throughput by reducing sample prep time 75%.

Toast Gastropub

June 2025 – August 2025

Line Cook

Richmond, VA

PROJECTS

Rubik's Cube Solver

Jan 2026 - Present

- Developing a robot using SO-101 arms to solve a Rubik's Cube
- Built a VLA and computer-vision model to accurately get state of Rubik's cube and output next action to solve
- Training arms via inverse reinforcement learning on simple policies (turn cube 90°, etc.)

Laundry Robot

Dec 2025

- Built an autonomous mobile system to collect scattered clothing end-to-end with ROS2.
- Developed a service to map any pixel on camera image into (x,y,z) coordinates using only an RGB-camera and AprilTags
- Utilized deep learning (YOLOv8) to identify clothing, a Catmull-Rom spline to trace a path, and calculated wheel velocities
- Implemented an Extended Kalman Filter for accurate localization, modeled covariance through experimentation

Wearable EMG Instrument (2nd Place - International IEEE Arduino Contest)

Aug 2024

- Designed op-amp circuitry to capture and process electrical impulses from the biceps
- Engineered a wearable instrument that generated musical notes based on muscle activation strength

LEADERSHIP

Hardware Lead

Jan - May 2025

iRobotics Micromouse, University of Illinois

Champaign, IL

- Led hardware design for maze-solving robots; ran weekly design reviews and taught PCB/PCBA fundamentals (15+ members).
- Fabricated custom PCBs in KiCAD with I2C/SPI sensor interfaces and analog filtering